

$$\begin{array}{ccc}
 V = \underbrace{\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_r\}}_{\mathcal{N}(\mathbf{T})^\perp = \mathcal{R}(\mathbf{T}^\dagger)} \oplus \underbrace{\text{Span}\{\mathbf{v}_{r+1}, \dots, \mathbf{v}_n\}}_{\mathcal{N}(\mathbf{T}) = \mathcal{R}(\mathbf{T}^\dagger)^\perp} & & \{\mathbf{0}_V\} \\
 \begin{array}{c} \textcolor{red}{\uparrow} \\ \textcolor{blue}{\downarrow} \end{array} \begin{array}{c} \textcolor{red}{\mathbf{T}^\dagger} \\ \textcolor{blue}{\mathbf{T}} \end{array} & \begin{array}{c} \textcolor{red}{\uparrow} \\ \textcolor{blue}{\downarrow} \end{array} \text{bijection} & \begin{array}{c} \textcolor{red}{\nearrow} \\ \textcolor{blue}{\searrow} \end{array} \\
 W = \underbrace{\text{Span}\{\mathbf{w}_1, \dots, \mathbf{w}_r\}}_{\mathcal{R}(\mathbf{T}) = \mathcal{N}(\mathbf{T}^\dagger)^\perp} \oplus \underbrace{\text{Span}\{\mathbf{w}_{r+1}, \dots, \mathbf{w}_m\}}_{\mathcal{R}(\mathbf{T})^\perp = \mathcal{N}(\mathbf{T}^\dagger)} & & \{\mathbf{0}_W\}
 \end{array}$$